

# Christi Sanderson

[christi.m.sanderson@cfl.rr.com](mailto:christi.m.sanderson@cfl.rr.com) | 321-438-4329

LinkedIn: <https://www.linkedin.com/in/christisanderson>

GitHub: <https://github.com/christi1>

## PROFESSIONAL SUMMARY

**Senior Full Stack Software Engineer (Java & AI Integration)** with 14+ years of experience at FedEx specializing in scalable, event-driven **Java Spring Boot microservices** architecture, high-throughput distributed systems, batch migrations, and legacy modernizations. Proven track record of delivering business-critical software for fast-paced **startups** and **enterprise** environments such as FedEx, Disney and Lockheed Martin. Independently expanded skills to full-stack design and development of advanced **AI/ML** technologies including observable multi-agent workflows (**LangGraph**), asynchronous orchestration (**PyTorch**), classic and deep machine learning (**Scikit-learn**, **TensorFlow**), **open-source LLMs**, **declarative** UIs, **RAG** pipelines, and scalable **Python** backends. Currently architecting hybrid serverless and containerized cloud topologies (**AWS** ECS, Lambda, SQS) emulated via **LocalStack** to modernize legacy applications into high-performance **Spring AI** environments.

## TECHNICAL SKILLS

- **AI & Orchestration:** LangGraph, LangChain, Multi-Agent Workflows, RAG, Ollama, Langfuse, Prompt Engineering, Copilot, Claude, VS Code
- **Core Machine Learning:** PyTorch, TensorFlow, Scikit-learn, MobileNetV2, Text-to-SQL
- **Data Architecture:** Oracle, Datastax Cassandra, Qdrant, ChromaDB, PostgreSQL, Pydantic, JSON Schema Validation
- **Backend Systems:** Java Spring Boot Microservices, Python, FastAPI, Node.js, REST APIs, JMS Asynchronous Messaging
- **Frontend & UI:** React, Angular, JavaScript/TypeScript, Tailwind CSS, Declarative UI, JSON Schema
- **Cloud & DevOps:** Docker, CI/CD Pipelines, GitHub, GCP, Azure, Cloud Foundry, Splunk, Dynatrace
- **Growing Skillset:** AWS (ECS, Lambda, SQS, S3, DynamoDB), LocalStack, Docker Compose, Testcontainers, Spring AI

## PROFESSIONAL EXPERIENCE

**FedEx Services** | Orlando, FL

*Full Stack Developer III* | January 2012 – June 2026

- **Awarded** prestigious **FedEx Bravo Zulu Awards** for above and beyond performance resulting in user interface performance improvements, revenue cost reductions, operational efficiencies, and enterprise PCI compliance and authentication solutions.
- **Designed** and built scalable, fault-tolerant **Java Spring Boot microservices** featuring service discovery and API gateway to execute concurrent batch processing across multiple global document repositories.
- **Enhanced** Enterprise Document Management core web services to enable client adoption resulting in **7% revenue growth** targeting an **\$8M revenue increase**.
- **Agile Team Lead** that **saved \$1.2M annually in licensing costs and increased operational efficiencies** by migrating vendor document repositories to the core FedEx platform.

- **Integrated AI tooling into production support**, accelerating systemic defect resolution and automating testing frameworks.
- **Developed Google Cloud Storage (GCS)** bulk export scripts to support compliance, audit research, and legal document reporting.
- **Engineered** React and Node.js data engines to visualize real-time system metrics, providing leadership with actionable performance dashboards.
- **Engineered** a high-throughput **Java** migration utility to safely transition 500M+ legacy documents into **Oracle, Datastax, and Google Cloud Storage (GCS)**.
- **Validated Datastax** conversion models by developing a Java Spring Boot **proof-of-concept (POC)** using embedded **Cassandra** databases with **Spring JPA**.
- **Achieved full PCI compliance** for high-volume global airway bill processing by partnering with InfoSec to implement enterprise eKMS encryption APIs and implementing custom SSL layers.
- **Modernized** legacy applications to **REST services** architecture while introducing **CI/CD automation** pipelines and metrics dashboards.

## PERSONAL AI/ML ENGINEERING PROJECTS

### Multi-Agent Research Approval Assistant

- **Lowered operational bottlenecks by 35%** using LangGraph stateful workflows with human-in-the-loop approvals and Langfuse observability in Docker for trace monitoring.
- Designed resilient, stateful, and resumable execution loops for advanced agent management.

### Enterprise Autonomous Analytics & Knowledge Engine

- **Achieved zero API costs, complete data privacy, and reduced hallucinations** by deploying a local Llama 3 engine with deterministic guardrails.
- Engineered a full-stack platform using decoupled microservices, Qdrant RAG pipelines, and PostgreSQL Text-to-SQL with React and Node.js frontends for deterministic UI layout rendering.

### OmniMind: Multi-Framework AI Agent Ecosystem

- **Improved multi-modal task routing speed by 50%** by building an asynchronous orchestration engine that unifies Scikit-learn, TensorFlow, and LangChain plus Ollama.
- Integrated tabular analytics and MobileNetV2 vision inference into a single scalable agent workflow.

### RAG Research Assistant

- **Reduced document retrieval time and user-perceived latency by 40%** by deploying a local LLM vector retrieval system and streaming token responses via Ollama and LangChain.
- Developed a full-stack general and semantic search chatbots backed by FastAPI.

### **Asynchronous DocIntel Pipeline: Hybrid Serverless & Event-Driven AI Engine** *(In Development)*

- **Architecting** a hybrid cloud topology combining **Serverless (AWS Lambda)** and **Containerized Microservices (AWS ECS)** to optimize cost and performance for a Java-based **Spring AI** application.
- **Simulating** the entire AWS ecosystem locally using **LocalStack** to mock live endpoints for **S3, SQS, SNS, Lambda, and DynamoDB** without incurring cloud infrastructure costs.
- **Isolating** microservices and local AWS mock services using custom, multi-tier **Docker Compose networks** to replicate production VPC security boundaries and DNS routing.
- **Decoupling** user-facing REST APIs from intensive LLM generation loops using an asynchronous **Event-Driven Architecture (EDA)** built on **Amazon SQS** and **Amazon SNS**.
- **Deploying** serverless **AWS Lambda** triggers to automatically ingest unstructured files from **Amazon S3**, extract text, and publish events to the processing queue.
- **Orchestrating** downstream multi-agent execution and embedding generation within containerized **AWS ECS** tasks to bypass serverless execution timeouts.
- **Automating** integration testing by leveraging **Testcontainers** in Java to programmatically spin up disposable **Qdrant** vector databases and LocalStack instances during CI/CD.

## **PREVIOUS PROFESSIONAL EXPERIENCE**

### **Walt Disney World** | Lake Buena Vista, FL

#### *Technical Specialist*

- **Developed** enterprise **Java** enhancements and data **workflows** for publish to core sales and booking composite system channels.
- **Engineered predictive recommendation** modules to process and store high-volume scoring data generated by Revenue Management engines to drive the strategic products and services offers presented for sale on the public internet.

### **M2 Systems Corporation (Muscato Corporation)** | Orlando, FL

#### *Senior Software Engineer*

- **Developed** multi-platform online transaction processing (OLTP) switch interfaces connecting disparate systems in healthcare and debit networks. Most notable clients included Blue Cross/Blue Shield and Sacred Heart Medical Center.
- **Collaborated with CEO** and product team during development at customer site.

### **TriCom, Inc. (ISR Global Telecom)** | Orlando, FL

#### *Embedded Systems Engineer*

- **Served** as a **founding** Embedded Systems Engineer in a 6-employee **spin-off** to **develop** line protocol task conversion software.
- **Deployed** independently to **Germany** to rollout production software and **execute** rigorous Acceptance Testing at the DaimlerChrysler corporate facility.

**Lockheed Martin Corporation** | Orlando, FL  
*Systems Engineer*

- **Selected for Engineering CoOp program** leveraging advanced mathematical modeling.
- **Developed** simulations to analyze navigation, tracking, and fixed-point scaling algorithms.
- **Commended** for **designing** a ground-based FLIR/LTT turret simulation to evaluate air-target feasibility.
- **Digitized** missile subsystem block diagrams (airframe, actuator, sensors, feedback loops) and tuned IMU filters via fly-out sensitivity and transfer alignment analyses.

## **CERTIFICATIONS & TRAINING**

**Udacity AI Tech Practitioner Training - Generative AI Fundamentals**  
Achieved Accomplished AI Solutions Architect Rating

**GitHub Copilot Training & AI Engineering Workshops**  
FedEx Corporate Training

**Mobile Application Development Certification**  
University of California, Irvine

## **EDUCATION**

**Bachelor of Science, Computer Science (Scientific Specialization), Cum Laude**

**Minor: Business Administration**

University of Central Florida, Orlando, FL  
Academic Scholarship, Time-Shortened Degree